

Recursion Subset, Subsequence

String Question.

Create a String with given String without 'a'.

Builder.

- Using an ans String in parameter.

str = baccad if (charAt[i] != 'a')
↓ (Add in to string)
b/accad Append(char)
↓ then check next char
b/ccad → bc/cad
↓
bced/

- Without passing any String Builder in parameter

(baccad)

[ch = b + " "]
↓

(accad)

[ch = " " +]
↓

(ccad)

[ch = "c" + " "]
↓

ch(cad)

Subsets :-

- * permutation & condition
- * ~~sub~~ subset $\Rightarrow$ Non-adjacent collection
 $\Rightarrow$ any possible selection of a ~~any~~ element from a set / array / list

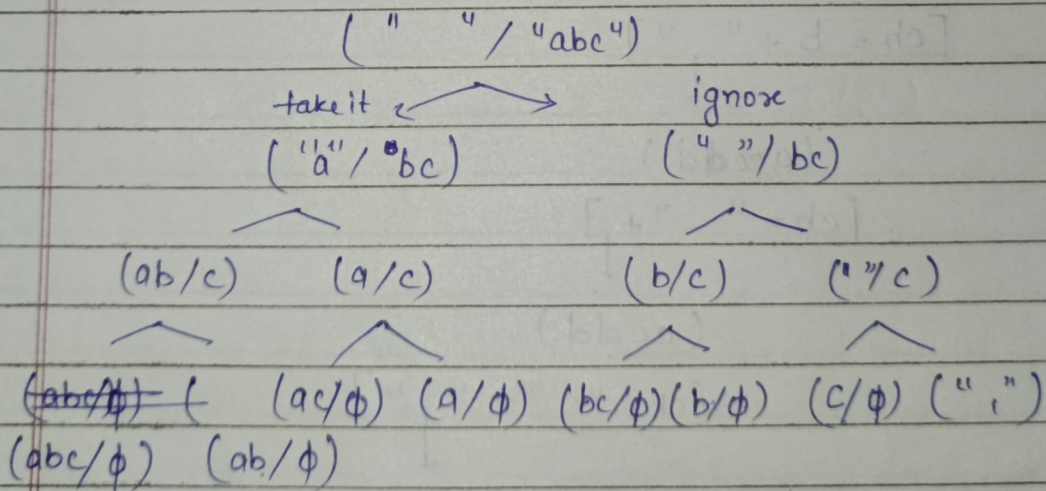
$[2, 3, 5] = [] ; [2] ; [5] ; [3] ; [2, 3] ; [3, 5] ; [5, 2] , [2, 3, 5]$

- Creating All Subsets

str = "abc"

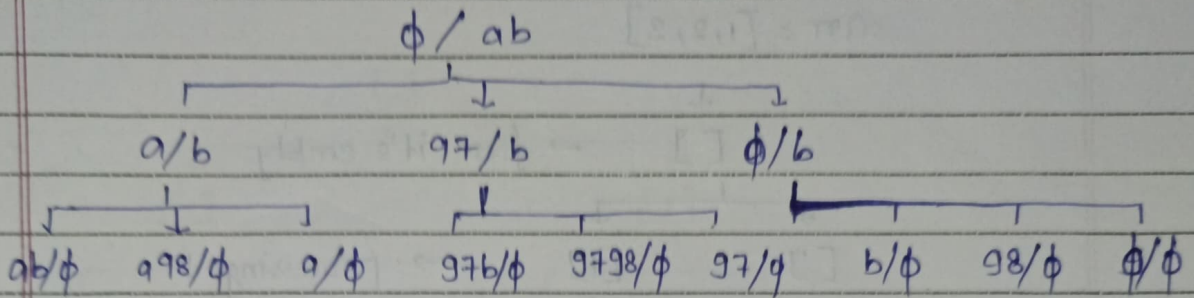
ans = ["", "a", "b", "c", "ab", "ac", "bc", "abc"]

- This is the pattern of taking some element & removing some if



ASCII value

str = "ab"

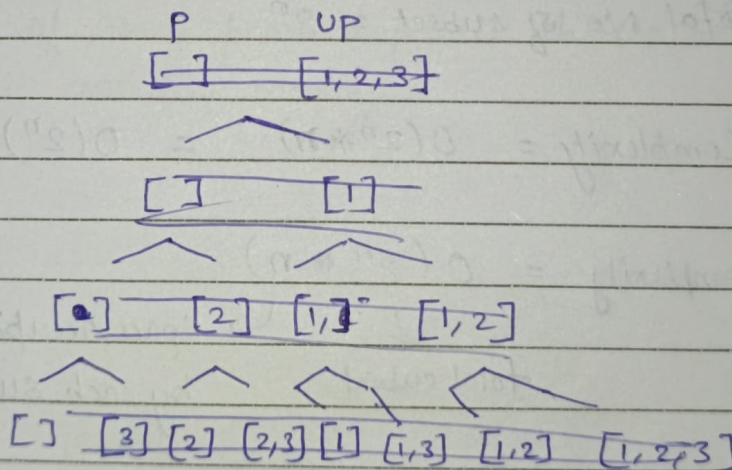


Find ASCII value

System.out.println("a" + 0); // 97

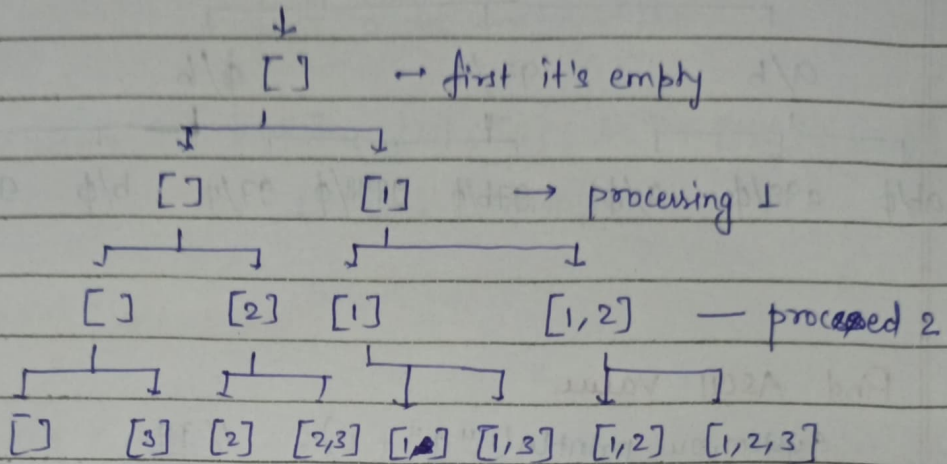
Iterations

arr = [1, 2, 3]



Iterations

arr = [1, 2, 3]



Complexity:-

Time taken at each level = n

Total No of subset = 2^n

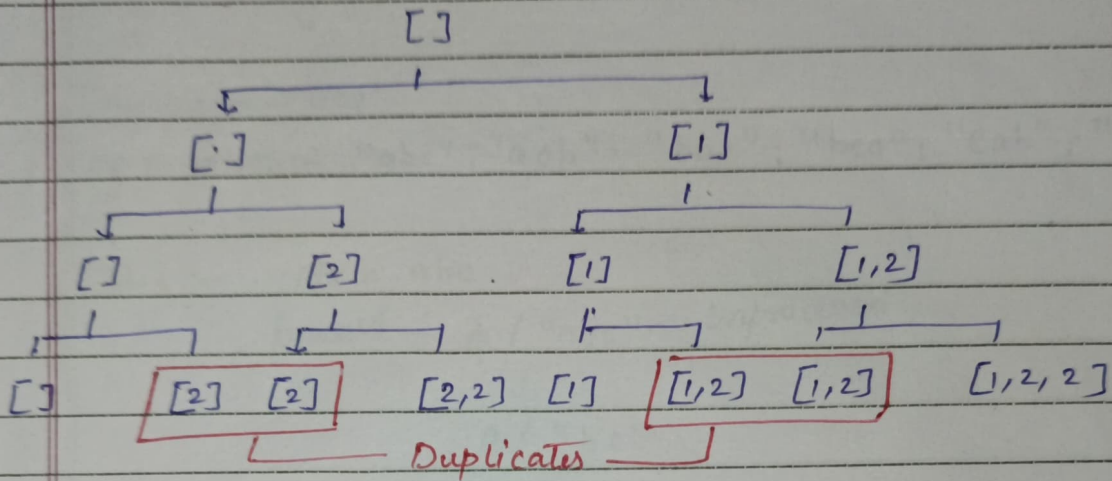
$$\text{Time Complexity} = O(2^n * n) = O(2^n)$$

$$\text{Space Complexity} = O(2^n * n)$$

$\downarrow$ $\hookrightarrow$
 total subset space occupied by each subsets

Duplicates

arr = [1, 2, 2]



* When we find duplicate element then add that element only in newly created subset.

* Because of above point duplicate should be together then 1st sort the array.